



Honda POS

Complete Honda POS System — Feature Documentation

- v1.0.12
- React 19 + Vite 7
- FIFO Batch Tracking
- Electron Desktop

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Target: Desktop POS for motorcycle parts & service retail with FIFO inventory costing & customer credit tracking

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Technical Specifications

§1 Point of Sale (POS) — Billing & Checkout

✓ Core POS Features

- **Real-time product search:** Search by name or category, filters live as you type
- **Barcode/QR scanner support:** USB HID scanner emulates keyboard input (plug & play, no drivers)
- **6-column product grid:** Display products with emoji/image, batch-linked selling price, and stock status (color-coded)
- **Shopping cart (persistent):** Cart items remain after completing a sale — you can keep selling OR clear manually with red button
- **Editable quantity & price:** Click quantity or price in cart to edit on-the-fly. Qty edits have stock limits. Price edits show strikethrough for original

📦 Batch-Aware Pricing (New)

- **Oldest-batch price display:** Product grid pulls selling price from the oldest available batch (FIFO order), not a fixed product price
- **Automatic fallback:** If no batch exists for a product, the product's base `sellingPrice` is used instead
- **Cart consistency:** When added to cart, each item locks in the batch-derived price at the moment of adding

💳 Payment & Checkout

- **Full payment (cash only):** Enter amount, automatic change calculation
- **Partial payment with customer credit:** If customer linked: pay less than total → remaining goes to customer's udhar. Error only shown if NO customer linked
- **Zero amount entry (full credit):** No amount entered + customer linked = full bill becomes udhar
- **Optional customer linking:** Button "Link to Customer (Udhar/Credit)" opens customer picker with favorites at top
- **Auto stock deduction:** Stock reduces after every sale instantly, consumed from the oldest batch first (FIFO)

🧾 Receipt Generation

- **Invoice design:** Shop name (custom font, always black & bold 900), Invoice badge, address, phone, date/time
- **Item table:** Description, Price, Qty, Total (with bold toggle for items)
- **Discount display:** Shows discount % and amount with bold toggle option

- **Custom labels:** "Credit (Udhar)" text is customizable in settings (can be "Baqi", "Remaining", anything)
- **Bold credit label option:** When ON: 17px font, 900 weight, red border separator (like total line). When OFF: 13px, normal weight
- **Terms & Conditions:** Optional T&C block at bottom with bold toggle
- **Footer message:** Custom shop sign-off (custom font, always bold 900, black)
- **Print button:** Opens print dialog, receipt formatted for thermal printer (80mm width)

Keyboard Shortcuts & Interactions

- **F11:** Toggle fullscreen mode (useful for kiosk setup)
- **Barcode input:** Green-bordered field auto-focuses, Enter-key triggers product add
- **Click quantity:** Turns into editable input (starts empty, shows current as placeholder)
- **Click price:** Turns into editable input for custom pricing
- **Escape key:** Cancels qty/price edits, reverts to previous value
- **Sliding cart:** ChevronRight in cart header hides it. Green cart button appears in search bar to show it again

§2 Products Management

Product CRUD & Organization

- **Add new products:** Name, barcode (QR/code39), category, unit (pieces/boxes/strips), emoji/image, cost price, selling price
- **Image upload:** Auto-resizes to 400px, compresses to JPEG 0.85 quality
- **Edit products:** Modify any field, changes apply instantly
- **Delete products:** With confirmation dialog (data safety)
- **Drag-and-drop reordering:** GripVertical handle visible on each product card, disabled during search/filter
- **Category management:** Create/edit/delete categories from "Manage Categories" modal
- **Stock display:** Color-coded: Green (>20), Orange (≤ 20), Red (0/out of stock)

Search & Filtering

- **Real-time search:** By product name or barcode
- **Category filter:** Dropdown to filter by category, "All" shows everything
- **Profit calculation:** Shows profit per unit = selling price – cost price. Visible in product detail modal
- **Barcode input field:** Scan to add product directly to cart without navigating to POS page

Data Persistence

- All products stored in `localStorage` under `pharma_products`
- Order preserved through drag-and-drop (saved immediately after reorder)
- Stock changes persist across sessions, kept in sync with batch quantities

§3 FIFO Batch Tracking (New in v1.0.12)

🌟 Why Batch Tracking

- Products purchased at different times often have different cost and selling prices
- FIFO (First-In-First-Out) ensures the oldest stock is sold first — matching real retail practice, especially for stock with limited shelf life
- Each sale's profit is calculated against the **actual cost** of the batch it was drawn from, not an average or a single fixed cost

📦 Adding a Batch

- **Batch fields:** Product, quantity, cost price, selling price, purchase date, optional expiry date, batch number, vendor name, notes
- **Auto stock sync:** Adding a batch automatically increases the linked product's total stock quantity
- **Multiple batches per product:** A single product can have many active batches queued in purchase order

📦 FIFO Consumption on Sale

- **Oldest-first logic:** When a sale is completed, quantity is deducted from the oldest batch with available stock first, then the next-oldest if the first runs out
- **Consumption record:** Each deduction is logged — batch ID, product ID, quantity taken, cost price at time of sale, selling price, linked sale ID
- **Accurate profit:** Profit per sale = $\sum (\text{selling price} - \text{actual batch cost price}) \times \text{qty consumed}$ from each batch

💰 Batch Selling Price on POS

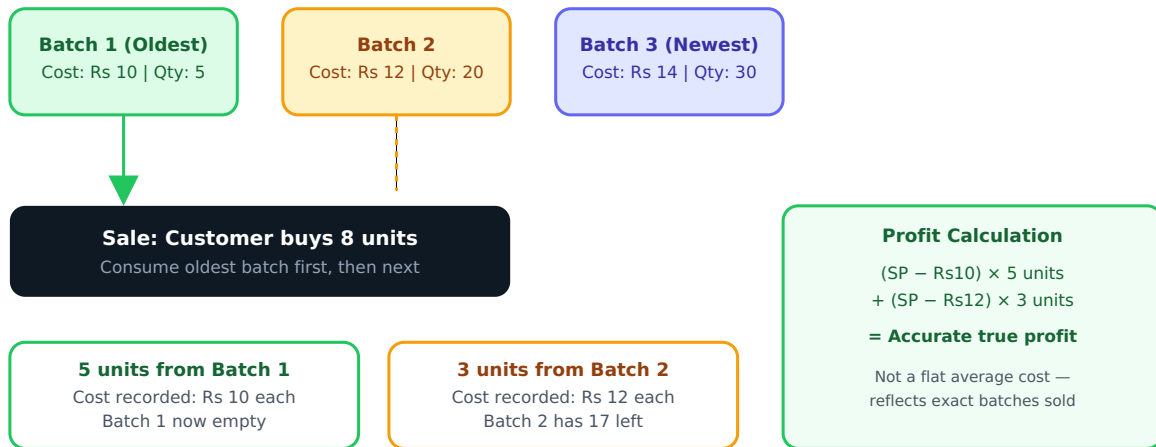
- The POS product grid displays the selling price of the **oldest available batch** — the one that will actually be sold next
- Falls back to the product's base selling price if no batch stock remains

⚠️ Batch Deletion & Archiving (Planned)

- Deleting a batch outright can break historical profit calculations if that batch was already sold from — its cost reference disappears from past sales
- Safer options under consideration: **archiving** a depleted/unwanted batch (hides it from active inventory but keeps it for profit history), or a **write-off** function to reduce quantity for damage/expiry while preserving the cost trail

- Only batches with zero remaining quantity and zero linked sales history are safe to fully delete

FIFO Batch Consumption Flow



Sale of 8 units draws first from the oldest batch, then spills into the next batch — each portion keeps its own recorded cost.

§4 Inventory Tracking

Stock Monitoring

- **Visual progress bars:** Green bar shows current stock as % of warehouse capacity
- **Stock levels:** Displays current qty, restock trigger threshold (configurable in settings)
- **Low stock alerts:** Items below threshold highlighted in orange/red (visual alert), with a popup shown on the first bill after opening the app
- **Color coding:** Green (sufficient), Yellow (approaching limit), Red (below threshold)

Restocking

- **Restock modal:** Enter quantity to add, optional purchase cost (for tracking spending)
- **Batch-based restocking:** Each restock can be entered as a new batch, preserving its own cost and selling price for FIFO
- **Auto-update:** Stock updates instantly across all pages (POS, reports, products, batches)

Inventory Reports

- **Stock summary:** Total items, items in stock, low stock count
- **Average stock value:** Total cost of inventory, calculated from live batch quantities
- **Stock turnover:** Items sold this month vs avg sell rate

\$5 Customer Management & Udhar (Credit) Tracking

🌟 Customer Features

- Complete customer profile system with purchase history & credit tracking
- Support for partial payments and manual debt addition
- Automatic pending amount calculation with formula transparency
- Print customer statements for record-keeping

👤 Customer List & Management

- **Customer profiles:** Name, phone, address (optional), favorite toggle
- **Favorite customers:** Quick access at top of list, visible in POS customer picker
- **Search & filter:** Search by name or phone number, filter favorites only
- **KPI row:** Total customers, customers with pending udhar, total pending amount (all customers)
- **Customer card displays:** Total spent, sales count, pending amount with color badge
- **Edit/Delete:** Modify customer details or remove from system (sales remain in reports)

💰 Pending Amount (Udhar) Calculation

- **Formula:** Pending = Total Billed – Amount Paid at Sale – Manual Payments + Manual Dues Added
- **Visual formula bar:** Shows exact calculation on customer detail page for transparency
- **Color-coded status:** Green (Clear), Yellow (<1000 pending), Red ($\geq$ 1000 pending)
- **Automatic calculation:** Recalculates after every payment/due entry
- **Display format:** Shows amount in currency or "✓ Clear" if zero

📄 Customer Detail View

- **Profile header:** Customer name, phone, address, customer since date, favorite star
- **KPI tiles (4):** Total Billed, Total Received (paid at sale + manual payments), Dues Added, Pending (Udhar)
- **Alert banner:** Red (if pending) or Green (if all paid) with action buttons
- **Two tabs:** Purchase History & Payment Ledger

📅 Purchase History Tab

- **All sales linked to customer:** Date, time, transaction ID, items purchased, total, amount paid,

status (Paid/Credit/Partial)

- **Discount display:** Shows discount amount if any
- **Due tracking:** Shows remaining due for each sale if not fully paid
- **Sorted by date:** Most recent first

Payment Ledger Tab (Combined)

- **Payment entries:** Money received from customer (minus icon, green)
- **Due entries:** Manual debts added (plus icon, orange)
- **Each entry shows:** Date, time, amount, type (Payment/Due), notes, delete button
- **Timeline view:** Sorted newest first, complete history of all transactions

+ Record Payment & Add Due Buttons

- **Record Payment:** Modal to enter amount received from customer, optional notes (e.g., "Paid full", "Partial")
- **Quick buttons:** "Full" button (auto-fill pending amount), "Half" button (auto-fill 50%)
- **Add Manual Due:** Modal to add debt not from a sale (old udhar, loan, etc.)
- **Both modals:** Have note field for documenting reason/reference
- **Mode switcher:** Tab switcher inside modal to toggle between Record Payment and Add Due

Print Customer Statement

- **Print button:** Top-right of customer detail page
- **Statement includes:** Customer name, phone, address, customer since date
- **Summary KPIs:** Total billed, total received, manual dues added
- **Outstanding balance box:** Red highlighted if pending, green highlighted if clear
- **Full tables:** All purchases (with due status), all payment/due entries
- **Print formatting:** A4 landscape, ready for direct print or PDF save

§6 Reports & Analytics

Overview Tab

- **Period selector:** Daily / Weekly / Monthly buttons to switch chart view
- **Product filter:** Dropdown to filter reports by specific product
- **Revenue chart:** Bar chart showing revenue for last 7 days (daily), 6 weeks (weekly), or 6 months (monthly)
- **Top Selling Products:** Horizontal progress bars showing units sold (top 6 products)
- **Revenue by Category:** Horizontal progress bars showing revenue per category (top 6)
- **Summary grid:** Total sales, gross revenue, total discounts, net revenue, avg order value, total items sold

FIFO-Accurate Profit

- Profit figures pull from the batch consumption log (`pharma_batch_consumptions`) rather than a single fixed cost per product
- Each sale's line item profit = (selling price – actual batch cost) summed across every batch portion consumed
- Gives a realistic view of margins when purchase costs fluctuate between restocks

Transactions Tab

- **Transaction list:** All sales as clickable cards showing TXN ID, date/time, items count, total, pending (if any)
- **Status badge:** Green checkmark if paid, red clock if unpaid/pending
- **Customer tag:** Shows customer name if linked (blue badge)
- **Action buttons:** Eye (view details), Edit, Print, Delete

Filtering & Sorting

- **Product filter:** Dropdown to show only sales of specific product
- **Date range:** "From" and "To" date pickers (calendar inputs)
- **Clear all filters:** Button to reset all filters at once
- **Count display:** Shows filtered transaction count at top

View Transaction Details

- **Modal popup:** Shows full invoice layout (shop name, address, phone, date/time, items table, totals)
- **Item details:** Product name, price per unit, quantity, line total
- **Calculation row:** Sub-total, discount, total, paid amount, change, credit (if any)
- **Action buttons:** Edit, Print, Delete right in the modal header

Edit Transaction

- **Edit fields:** Discount % and Amount Paid only (items cannot be modified)
- **Recalculation:** Totals and change auto-calculate as you edit
- **Note:** Stock levels are NOT restored when editing — manual restock if needed

Export & Reset

- **Export CSV (green button):** Download all transactions as CSV file (TXN ID, date, time, qty, amounts)
- **Reset Reports:** Permanently delete ALL sales data (requires double confirmation)

§7 Settings, Backup & Restore

Shop Information

- **Shop logo:** Uploadable, auto-resized, printed above shop name on receipt
- **Shop name:** Displayed on all receipts, settings, POS header
- **Phone number:** Shown on receipts
- **Address:** Shown on receipts with bold toggle option
- **GSTIN & currency:** Configurable tax ID and currency symbol

Receipt Formatting & Fonts

- **Shop name font:** Dropdown (Pacifico, Lobster, Great Vibes, Playfair Display, Poppins, Bebas Neue, Dancing Script, Caveat)
- **Body font:** Dropdown for item descriptions and text
- **Footer font:** Dropdown for shop sign-off message
- **Footer text:** Custom message printed at bottom of receipt (e.g., "Thank You!", shop tagline)
- **Terms & Conditions:** Multi-line text field, printed at bottom of receipt

Thermal Printer Boldness (Receipt Bold Options)

- **Bold Address / Phone / Date & Time / Discount / Terms:** Independent toggles for each receipt section
- **Bold Credit/Udhar Label:** When ON: 17px font, 900 weight, red border. When OFF: 13px, normal weight
- **Bold Item Rows:** When ON: product name, price, qty, total all print bold and black
- **Note:** Shop name and footer are ALWAYS bold & black automatically

POS Settings

- **Enable Discount:** Toggle to show/hide discount selector at checkout
- **Enable GST:** Toggle to include/exclude GST calculation in totals
- **Cart font size:** Dropdown (Small, Medium, Large, Extra Large) with live preview
- **Low stock threshold & alert toggle:** Configure trigger quantity and enable/disable the pop-up

Credit Label Customization

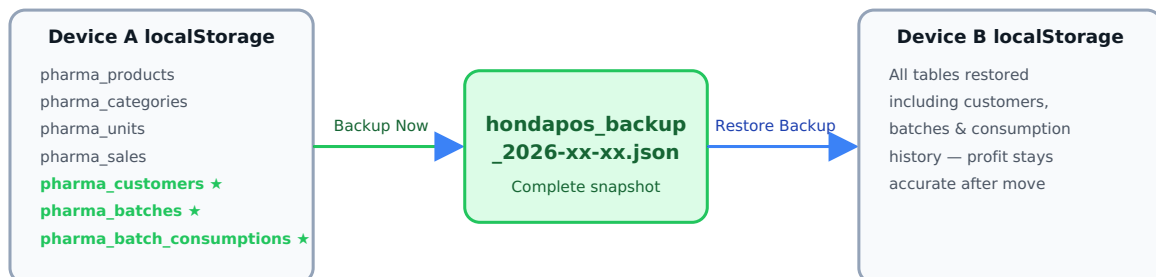
- **Credit/Udhar Label:** Text input field to customize the label on receipts

- **Default:** "Credit (Udhar)"
- **Examples:** Can be "Baqi", "Baki Raham", "Remaining", anything you prefer
- **Applied to:** All receipts (printed, modal, PDF) and customer statement

📁 Backup & Restore — Full Data Coverage (Updated)

- **Backup Now (JSON):** Downloads a complete snapshot of every data table in one file
- **Included in backup:** Products, Categories, Units, Sales, Customers, **Batches, Batch Consumptions**, and Settings
- **Restore Backup:** Upload a previous backup file to replace the entire local database — a two-step confirmation prevents accidental data loss
- **Cross-device migration:** Backup on one PC, restore on another — batches and customer/udhar records now travel with the file (previously only products/sales/settings were included)
- **Export to Excel:** Creates a multi-sheet .xlsx file (Products, Sales, Sale Items, Categories, Units, Settings) for accountants
- **Note:** All data lives in browser localStorage inside the Electron app's AppData folder — backups are the only way to move data between machines

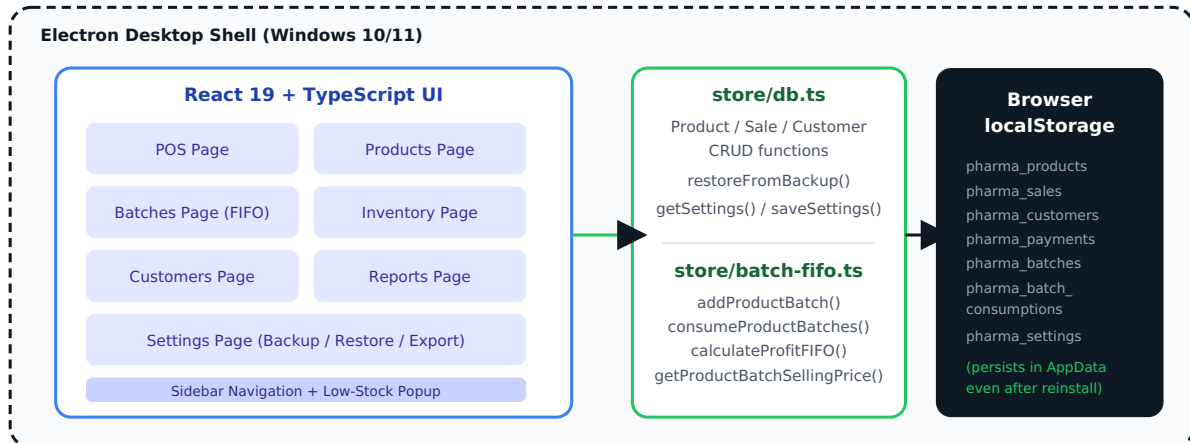
Backup & Restore Data Flow



★ marks the tables added in this update — customer/udhar records and FIFO batch data now included in every backup and restore cycle.

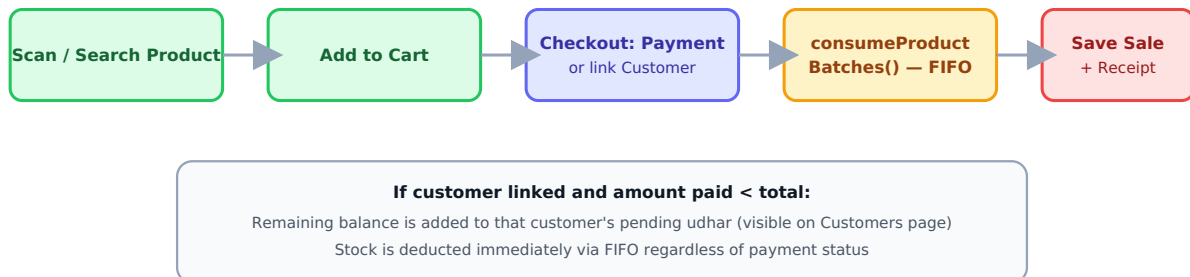
§8 System Architecture & Data Flow

Application Architecture



React pages call into two store modules — general CRUD (db.ts) and FIFO batch logic (batch-fifo.ts) — both persisting to browser localStorage inside the Electron shell.

Sale Transaction Pipeline



Every completed sale triggers FIFO batch consumption and, if under-paid with a linked customer, an automatic udhar entry.

Project Structure

- **src/App.tsx**: Root component with page routing (adds 'batches' route)
- **src/pages/**: POSPage, ProductsPage, BatchesPage, InventoryPage, ReportsPage, SettingsPage, CustomersPage, CustomerDetailPage
- **src/components/**: Sidebar navigation with collapsible menu
- **src/store/db.ts**: Core database layer — products, sales, customers, payments, settings, backup/restore
- **src/store/batch-fifo.ts**: FIFO batch engine — addProductBatch, consumeProductBatches, calculateProfitFIFO, getProductBatchSellingPrice
- **electron/main.cjs**: Electron main process (F11 fullscreen, focus fix)

§9 Technical Specifications

Tech Stack

- **Frontend:** React 19, TypeScript, Vite 7
- **Styling:** TailwindCSS 4, custom CSS
- **Charts:** Recharts (bar, line, pie charts)
- **Icons:** Lucide React
- **Data:** Browser localStorage (localStorage API)
- **Desktop:** Electron 33 (Windows 10/11)
- **Build:** vite-plugin-singlefile (single HTML output), electron-builder for .exe

Database Schema (localStorage) — Updated

Key	Contents
pharma_products	Array of products with name, price, stock, category, unit
pharma_categories	Category list for organization
pharma_units	Unit types (pieces, boxes, strips, etc.)
pharma_sales	Complete transaction history with customer linking
pharma_customers	Customer profiles with contact info & favorites
pharma_payments	Payment and manual debt entries linked to customers
pharma_batches	(New) Purchase batches — cost, selling price, qty available, expiry, vendor
pharma_batch_consumptions	(New) Log of every FIFO deduction — batch, qty, cost, sale ID
pharma_settings	User preferences, fonts, toggles, shop info

Note: these are internal localStorage key names carried over from the original codebase — technical identifiers only, unrelated to the Honda POS branding shown throughout the app.

Key Batch Interfaces

Interface	Fields
ProductBatch	id, productId, costPrice, sellingPrice, qty, qtyAvailable, purchaseDate, expiryDate, batchNumber, vendorName, notes, createdAt
BatchConsumption	batchId, productId, qty, costPrice, sellingPrice, saleId, consumedAt

UI/UX Design System

- **Brand color:** #ef4444 / #22c55e accents depending on module

- **Dark sidebar:** #0f1923 with gray text
- **Page background:** #f8fafc (light gray)
- **Cards:** White with 1px #e2e8f0 border, subtle shadows
- **Icons:** Lucide React (consistent 16-20px sizes)
- **Buttons:** Green for primary (CTA), white for secondary, red for destructive
- **Modals:** Dark overlay (bg-black/50) with backdrop blur
- **Animations:** 200-300ms transitions for smooth interactions

📦 Build & Deployment

- **Development:** `npm run electron:dev` (Vite dev server + Electron window)
- **Build:** `npm run build` → single index.html in dist/
- **Desktop installer:** `npm run dist` → `release/Honda POS Setup 1.0.0.exe`
- **CSP dev warning:** Electron's "Insecure Content-Security-Policy" console warning appears only in development mode — it does not affect the packaged production build
- **Target platform:** Windows 10/11 desktop, offline-first
- **Offline:** 100% offline-first, no internet required

☆☆ Complete Feature Summary

8

Complete Pages

FIFO

Batch Costing Engine

100%

Offline Capable

✓ Full POS with barcode scanning & real-time search

✓ Batch-aware pricing pulled from oldest available batch

✓ FIFO stock consumption with per-batch cost tracking

✓ Accurate profit calculated from real batch costs

✓ Customer credit (Udhar) tracking with formula

✓ Partial payments & manual debt management

✓ Complete product CRUD with drag reordering

✓ Inventory with low-stock alerts & restocking

✓ Reports: Period filters (daily/weekly/monthly)

✓ Transaction edit, delete, print, view details

✓ Print customer statements with full ledger

✓ Custom fonts & 8 bold toggles for receipts

✓ Custom credit label (Baqi, Udhar, etc.)

✓ Full-coverage JSON backup — products, sales, customers, batches

✓ Cross-device restore with zero data loss

✓ Electron desktop app with .exe installer

🔑 Key Advantages

- **No setup required:** Works offline, no database installation needed

- **Real-world costing:** FIFO batches mean profit reports reflect what stock actually cost
- **Low hardware footprint:** ~4GB RAM, any i3 processor, single-file deployment
- **Data safety:** Complete backup/restore in one click, covering every table including customers and batches
- **Professional receipts:** Custom fonts, colors, bold options — matches your branding
- **Modern UI:** Clean design, fast, responsive on any screen size
- **Pakistan-ready:** Support for Urdu-style labels, PKR currency, thermal printers (via Electron)



Honda POS v1.0.12

Complete Honda POS System with FIFO Batch Tracking

✓ **FIFO Costing** • ✓ **Customer Credit Tracking** • ✓ **Full Backup Coverage** • ✓ **100% Offline** • ✓ **Desktop Ready**